

EPPS 6354 Information Management

Assignment 5

Choose 1 or 2 to answer and answer the rest.

Prepare your answers in presentation format and be ready to present in class

1. An E-R diagram can be viewed as a graph. What do the following mean in terms of the structure of an enterprise schema?
 - a) The graph is disconnected.
 - b) The graph has a cycle.
2. Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted (Hint: use miro.com to draw the diagram with relationship sets).
3. We can convert any weak entity set to a strong entity set by simply adding appropriate attributes. Why, then, do we have weak entity sets?
4. SQL exercise:
 - a) Consider the employee database

```
employee (ID, person_name, street, city)
works (ID, company_name, salary)
company (company_name, city)
manages (ID, manager_id)
```

1.

where the primary keys are underlined. Give an expression in SQL for each of the following queries. (Hint: use **from** *employee as e*, *works as w*, *company as c*, *manages as m*)

- i. Find ID and name of each employee who lives in the same city as the location of the company for which the employee works.
 - ii. Find ID and name of each employee who lives in the same city and on the same street as does her or his manager.
 - iii. Find ID and name of each employee who earns more than the average salary of all employees of her or his company.
- b) Consider the following SQL query that seeks to find a list of titles of all courses taught in Spring 2017 along with the name of the instructor.
- ```
select name, title
from instructor natural join teaches natural join section natural join course
where semester = 'Spring' and year = 2017
```

What is wrong with this query? (Hint: check book website)